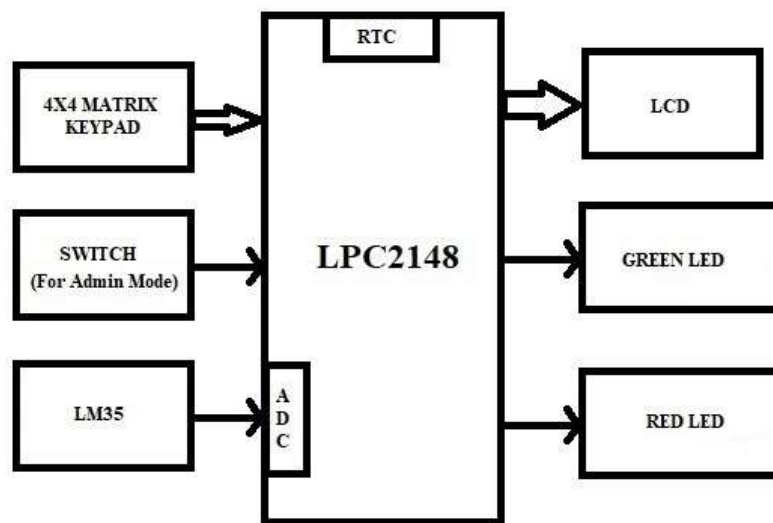


EVENT BOARD: RTC-DRIVEN MESSAGE DISPLAY SYSTEM

OBJECTIVE:

The main objective of **Event Board** is to automatically display predefined messages on an LCD with scrolling mechanism at specific times using the RTC in LPC2148 and allow secure access to **edit time and select messages** through an admin mode using a switch, keypad, and password protection.

BLOCK DIAGRAM:



REQUIREMENTS:

HARDWARE REQUIREMENTS:

- LPC2148
- LCD
- KEYPAD
- LED'S
- LM35
- BUZZER

SOFTWARE REQUIREMENTS:

- PROGRAMMING IN EMBEDDED C
- KEIL C COMPILER
- FLASH MAGIC

PROJECT WORKFLOW:

The **EventBoard: RTC-Driven Message Display System** using **LPC2148** is a real-time, automated message display system that shows **predefined messages** on a 16x2 LCD with scrolling mechanism at specific times using the on-chip RTC.

The system has **10 fixed messages** stored in controller memory and allows users to securely enter **Admin Mode** using a switch and a **password-protected keypad**.

In Admin Mode, the user can **edit the current time** and **select which messages** should be active for the day. During normal operation, if a scheduled message matches the RTC time, it is displayed and a **green LED is turned ON** to indicate active schedule display. If no message is scheduled, the system instead shows the **real-time clock** and **room temperature** (measured via an **LM35 sensor** using the LPC2148's **on-chip ADC**), and a **red LED is turned ON** to indicate idle display mode.

This project combines secure access, RTC-based control and environmental monitoring making it a complete embedded system application.

Note: Initially all messages are selected. FYI

Supported Information for programming:

```
#define TOTAL_MESSAGES 10

typedef struct {

    u8 hour;

    u8 minute;

    char text[80];

    u8 enabled;    // 1 = show, 0 = skip (user controlled)

} Message;
```

```
Message messageList[TOTAL_MESSAGES] = {  
    {7, 45, "Good Morning! Classes Start Soon", 1},  
    {13, 45, "C Programming Session in Classroom number 2", 1},  
    {10, 15, "C module theory exam in 4th floor lab1", 1},  
    {10, 15, "C module lab exam in 4th floor lab2 and middle lab at 10:30AM", 1},  
    {12, 45, "Lunch Break from 1PM – 2PM", 1},  
    {9, 45, "ARM Workshop on external interrupts in LAB1 at 10AM", 1},  
    {9, 45, "ARM kit issue time from 10AM – 10:30AM in middle lab", 1},  
    {15, 15, "Only 15 mins break time for next ARM session", 1},  
    {17, 00, "Revise today's Class programs at home!", 1},  
    {17, 45, "End of Day – See You Tomorrow!", 1}  
};
```

*****ALL THE BEST*****

Note: Develop the RTC time editing feature in a fully user-interactive manner.